Shahjalal University of Science and Technology

Department of Computer Science and Engineering

4th Year 1st Semester Final Examination, June 2024 (Session: 2019-20)

Course Code: CSE 475 Course Title: Machine Learning

Time: 3.0 hrs **Credits: 3.0** **Total Marks: 100**

Group A

[Answer all questions]

1. Answer any FIVE. **5x2 = 10**

a) What is plurality voting?

b) Name two loss functions.

c) What is hyperparameter? Give examples.

d) What is meant by a bootstrap sample?

e) Give two examples of reinforcement learning.

f) In which case, the entropy is zero in decision tree algorithm?

g) What is meant by high bias?

2. Answer any FOUR. **4x5 = 20**

a) Explain the meaning of the following piece of Python code.

```
model.compile(optimizer="rmsprop",
              loss="binary_crossentropy",
              metrics=["accuracy"])
model.fit(x_train, y_train, epochs=4, batch_size=512)
```

b) What is the main limitation of perceptron algorithm for classification? How is it resolved with the logistic regression method? Explain.

c) Why is dimensionality reduction required in ML algorithms? Discuss the PCA technique to achieve data compression.

d) Define the term TF-IDF. How does TF-IDF assess the word relevancy in encoding?

e) You are asked to determine the most discriminative feature in a dataset. How will you do it?

f) How is the nonlinearly separable case dealt using slack variables in SVM? Explain.

2x10 = 20

f) How is the nonlinearly separable case dealt using slack variables in SVM? Explain.

3. Answer any TWO. **2x10 = 20**

a) Suppose you have written a classifier for email spam filtering. You have tested your classifier with some data and got the following predictions. Compute the confusion matrix from the output and calculate **Accuracy**, **Sensitivity**, **Precision, and f1 score** from that matrix **3 + 7**

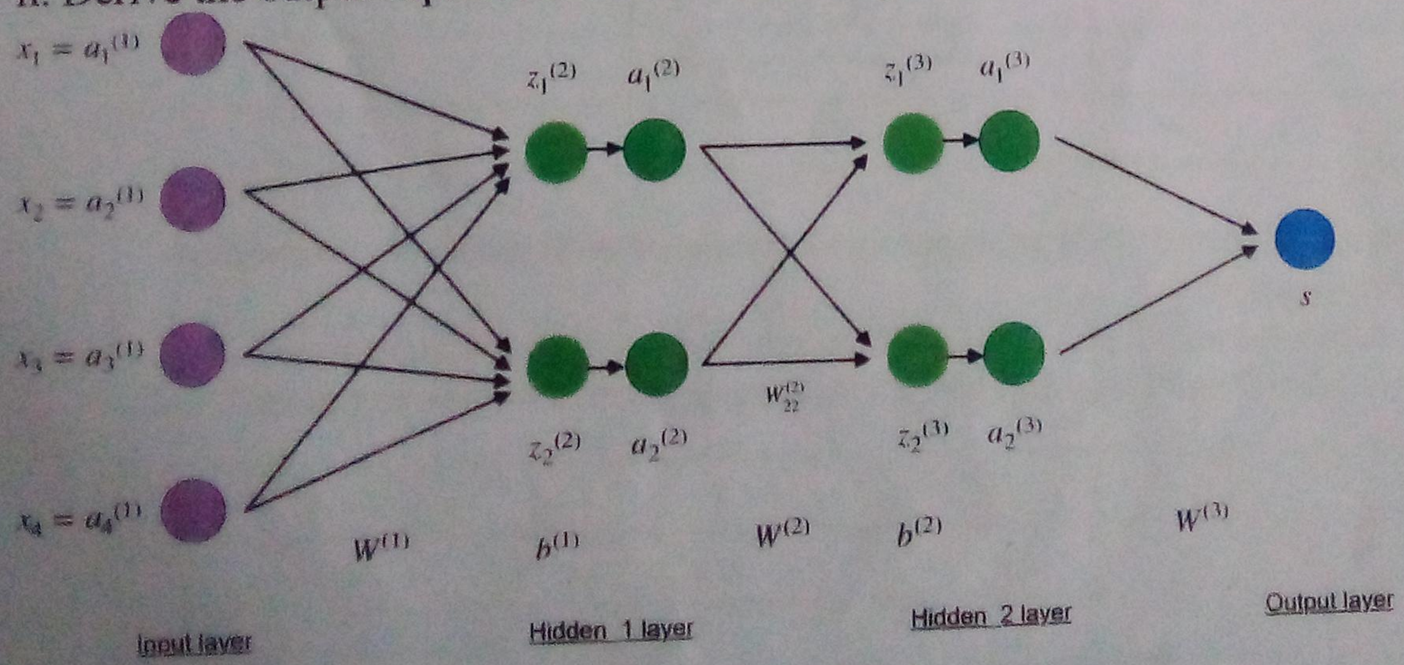
SL.	Target	Prediction
1	Spam	Not Spam
2	Spam	Spam
3	Not Spam	Not Spam
4	Spam	Spam
5	Not Spam	Spam
6	Not Spam	Not Spam
7	Not Spam	Not Spam
8	Spam	Spam
9	Spam	Not Spam
10	Not Spam	Not Spam

b) i. What is the importance of feature scaling for optimal performance? Explain the standardization technique of feature scaling. Name one ML technique which is not impacted by feature scaling. **6 + 4**

ii. Explain how L1 regularization is used for feature selection.

c) i. Briefly explain the steps for training a neural network. **5 + 5**

ii. Derive the output expression **S** for the following network.



Group B

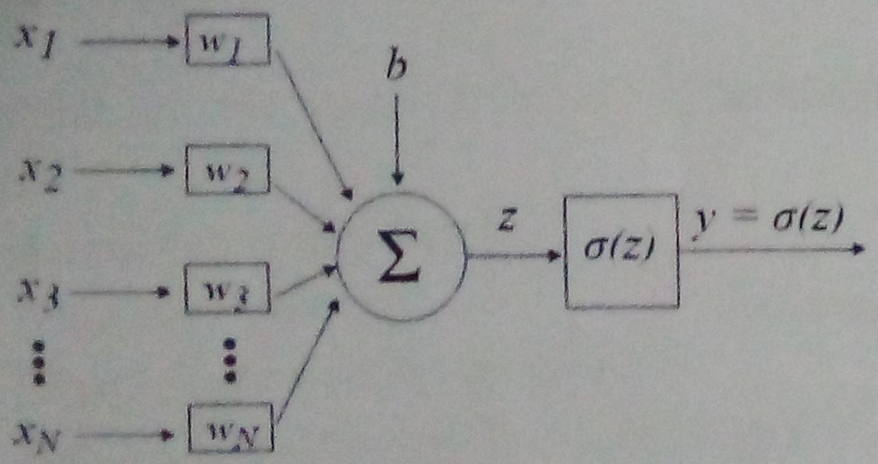
[Answer all questions]

4. Answer any FIVE. **5x2 = 10**

a) What is the main limitation of ensemble learning?
b) Give example of a nominal and an ordinal feature.
c) How does batch gradient descent differ from SGD?
d) Why should the activation function in ANN be differentiable?
e) Why is KNN called a lazy learner?
f) What are outliers?
g) Which evaluation metrics are good for imbalanced data?

5. Answer any FOUR. **4x5 = 20**

a) Define the purpose of each of the variables in the following figure.



b) Explain with a figure how a model selection is achieved together with tuning hyperparameters using nested k-fold cross-validation.
c) Show with an example how one-hot encoding is performed on a nominal feature? What is the main disadvantage of this approach?
d) How does K-Means clustering works? Draw the flowchart of K-Means clustering.
e) What is the purpose of max-pooling in CNN? Illustrate the process with an example.
f) Why KNN is termed as a non-parametric method? What happens when the majority voting in KNN results in a tie?

6. Answer any TWO. **2x10 = 20**

a) i. For the following dataset with 3 features (X, Y, Z) and 2 classes (I, II), Which attribute to split first according to entropy in decision tree algorithm? **6 + 4**



b) Explain with a figure how a model selection is achieved together with tuning hyperparameters using nested k-fold cross-validation.

c) Show with an example how one-hot encoding is performed on a nominal feature? What is the main disadvantage of this approach?

d) How does K-Means clustering works? Draw the flowchart of K-Means clustering.

e) What is the purpose of max-pooling in CNN? Illustrate the process with an example.

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2x10 = 20

6. Answer any TWO.

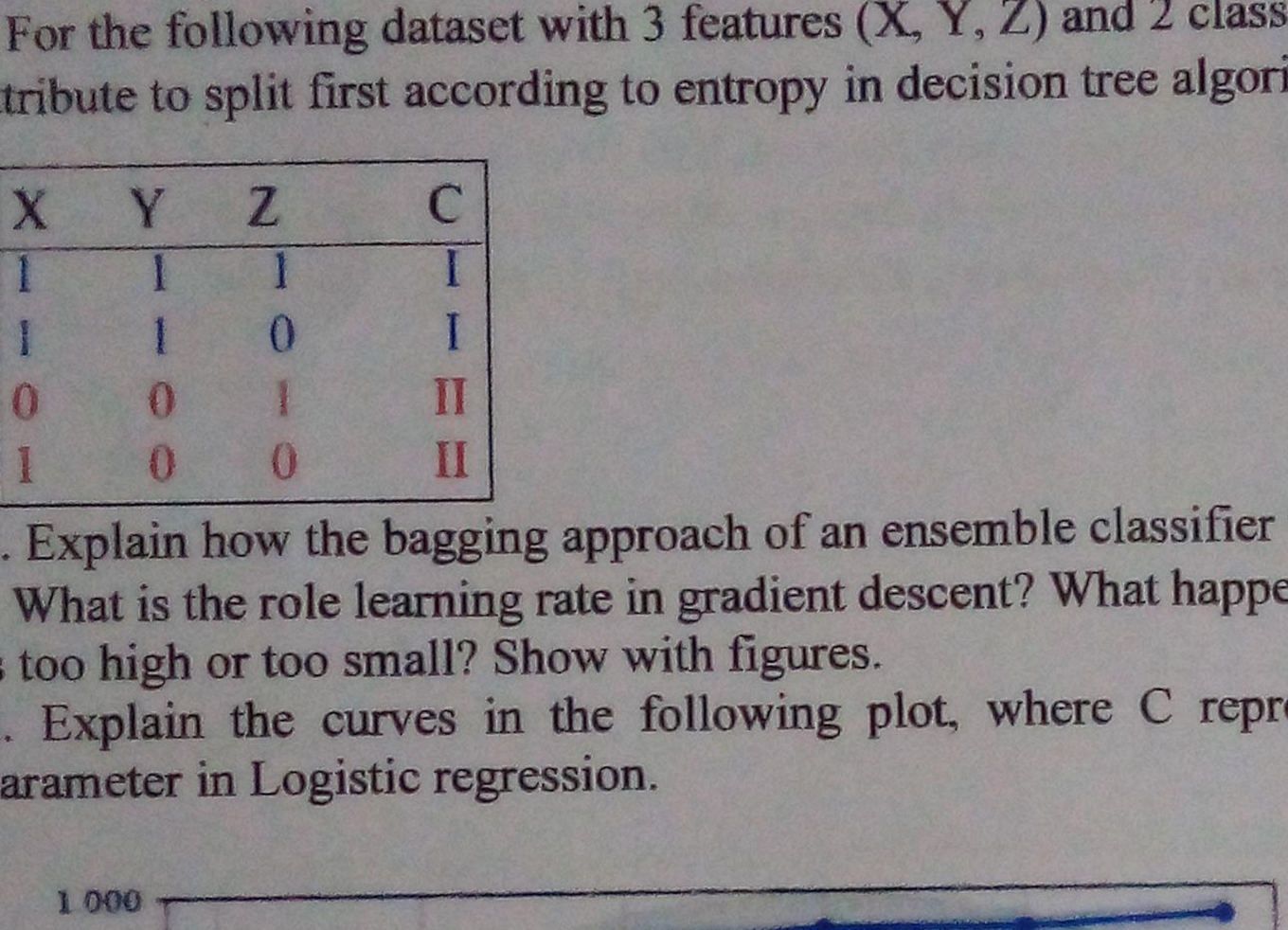
a) i. For the following dataset with 3 features (X, Y, Z) and 2 classes (I, II), Which attribute to split first according to entropy in decision tree algorithm? **6 + 4**

X	Y	Z	C
1	1	1	I
1	1	0	I
0	0	1	II
1	0	0	II

ii. Explain how the bagging approach of an ensemble classifier works?

b) i. What is the role learning rate in gradient descent? What happens when the learning rate is too high or too small? Show with figures. **5 + 5**

ii. Explain the curves in the following plot, where C represents the regularization parameter in Logistic regression.



c) i. What key characteristic gives CNN an upper hand compared with a densely connected layer in image classification? **3 + 7**

ii. Describe with how a kernel of size 3×3 is convolved with an input feature map of size 5×5 with stride 2×2 and input depth = 1. How many patches will be extracted?